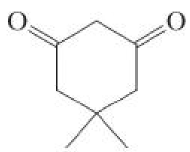
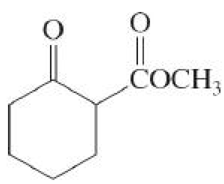
Problems

27. Arrange the following substances in order of increasing acidity. Estimate pKapK_a values for each.

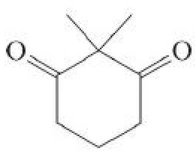
a. 

b. CH3CO2HCH_3CO_2H

c. CH3OHCH_3OH

d. 

e. CH3CHOCH_3CHO

f. 

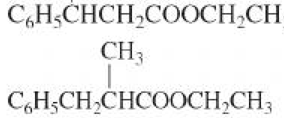
g. CH3O2CCH2CO2CH3$CH_3O_2CCH_2CO_2CH_3$

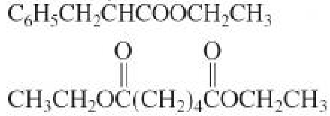
h. CH3O2CCO2CH3$CH_3O_2CCO_2CH_3$

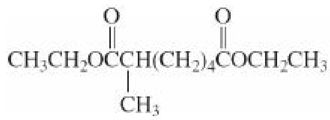
28. Give the expected results of the reaction of each of the following molecules (or combinations of molecules) with excess NaOCH2CH3 $NaOCH_2CH_3$ in CH3CH2OH,CH_3CH_2OH, followed by aqueous acidic work-up.

a. CH3CH2CH2COOCH2CH3$CH_3CH_2CH_2COOCH_2CH_3$

b. 

c. 

d. 

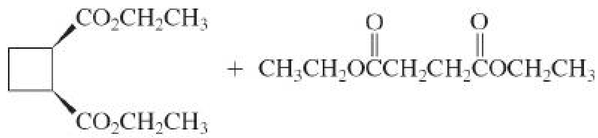
e. 

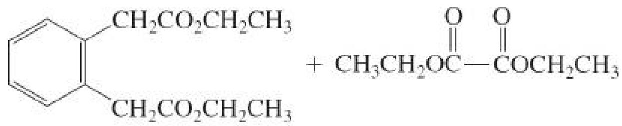
f. C6H5CH2CO2CH2CH3+HCO2CH2CH3

$C_6H_5CH_2CO_2CH_2CH_3 + HCO_2CH_2CH_3$

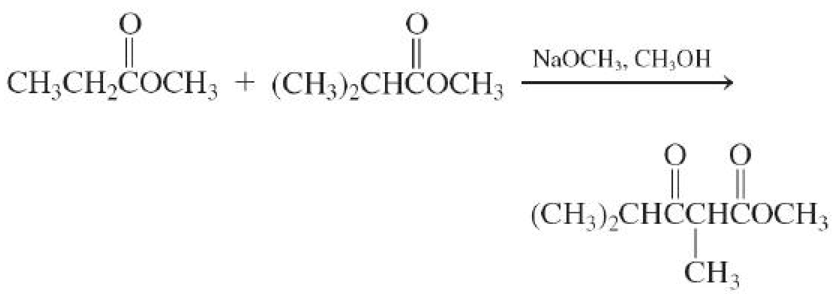
g. C6H5CO2CH2CH3+CH3CH2CH2CO2CH2CH3

$C_6H_5CO_2CH_2CH_3 + CH_3CH_2CH_2CO_2CH_2CH_3$

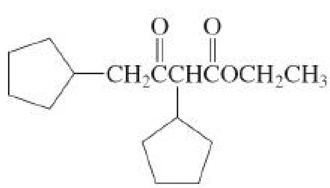
h. 

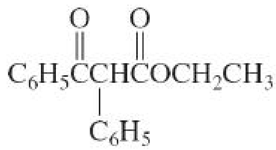
i. 

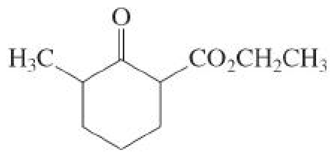
29. The following mixed Claisen condensation works best when one of the starting materials is present in large excess. Which of the two starting materials should be present in excess? Why? What side reaction will compete if the reagents are used in comparable amounts?

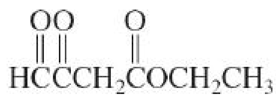


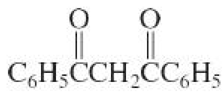
30. Suggest a synthesis of each of the following ββ-dicarbonyl compounds by Claisen or Dieckmann condensations.

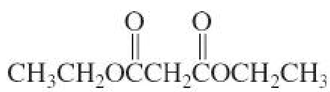
a. 

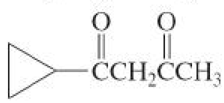
b. 

c. 

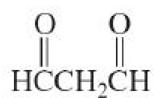
d. 

e. 

f. 

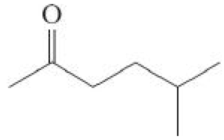
g. 

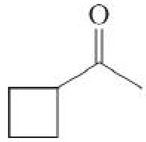
31. Do you think that propanedial, shown below, can be easily prepared by a simple Claisen condensation? Why or why not?

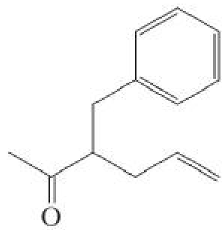


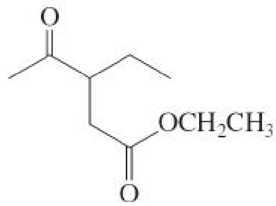
Propanedial

32. Devise a preparation of each of the following ketones by using the acetoacetic ester synthesis.

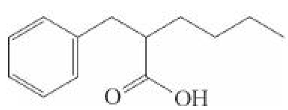
a. 

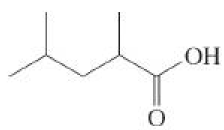
b. 

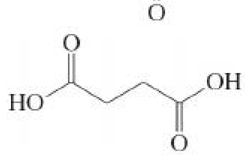
c. 

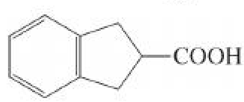
d. 

33. Devise a synthesis for each of the following four structures by using the malonic ester synthesis.

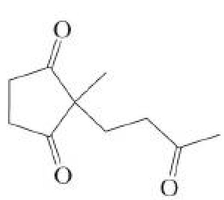
a. 

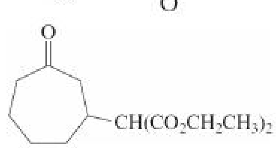
b. 

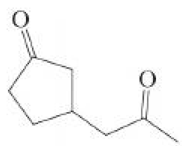
c. 

d. 

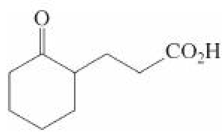
34. Use the methods described in Section 23-3, with other reactions if necessary, to synthesize each of the following targets. In each case, your starting materials should include one aldehyde or ketone and one β-dicarbonyl compound.

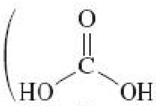
a. 

b. 

c. 

[**Hint:** Decarboxylations are necessary for parts (c) and (d).]

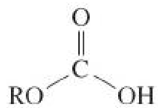
d. 

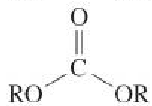
35. **CHALLENGE** Carbonic acid, H2CO3,H_2CO_3, (), is commonly assumed to be an unstable species that easily decomposes into a molecule of water and a molecule of carbon dioxide: H2CO3→H2O+CO2↑.$H_2CO_3 \rightarrow H_2O + CO_2 \uparrow.$ Indeed, the evidence of our own experience in opening a container of any carbonated beverage supports this apparently obvious notion. However, in 2000 it was discovered that this assumption is not quite correct: Carbonic acid is actually perfectly stable and isolable in the complete absence of water. Its decomposition, which is a decarboxylation reaction, is strongly catalyzed by water. It is exceedingly difficult to completely exclude moisture without the use of special techniques, which explains why carbonic acid has been such an elusive species to obtain in pure form.

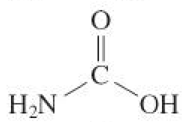
> How strong is carbonic acid? The pKa$\mathrm{p}K_\mathrm{a}$ value usually quoted, about 6.4, is misleading because it refers to the acidity of the equilibrium mixture of aqueous CO2CO_2 and H2CO3,H_2CO_3, which contains mostly molecular CO2.CO_2. The true acidity of carbonic acid is much greater, as confirmed by a study in 2009. It places the pKa$\mathrm{p}K_\mathrm{a}$ of H2CO3H_2CO_3 at around 3.5, close to that of formic acid (3.6).

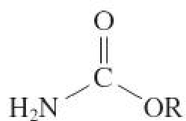
Based on the discussion of the mechanism of decarboxylation of 3-ketocarboxylic acids in Section 23-2, suggest a role for a molecule of water to play in catalyzing the decarboxylation of carbonic acid. **(Hint:** Try to arrange one water and one carbonic acid molecule to give a six-membered ring stabilized by hydrogen bonds, and then see if a cyclic aromatic transition state for decarboxylation exists.)

36. Based on your answer to Problem 35, predict whether or not water should catalyze the decarboxylation of each of the following compounds. For each case in which your answer is yes, draw the transition state and the final products.

a.  (alkyl carbonate: a monoester of carbonic acid)

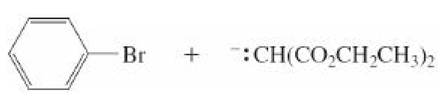
b.  (dialkyl carbonate: a diester of carbonic acid)

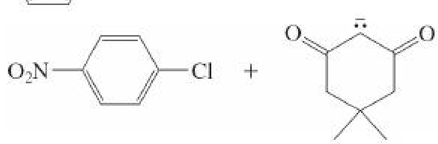
c.  (carbamic acid)

d.  (a carbamate ester)

37. Write out, in full detail, the mechanism of the Michael addition of malonic ester to 3-buten-2-one in the presence of ethoxide ion. Be sure to indicate all steps that are reversible. Does the overall reaction appear to be exo- or endothermic? Explain why only a catalytic amount of base is necessary.

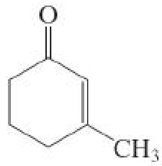
38. Give the likely products of each of the following reactions. All are carried out in the presence of a Pd catalyst, a ligand for the metal, such as a phosphine, and heat. (**Hint:** Refer to Section 22-4 for related Pd-catalyzed reactions of halobenzenes with a nucleophile.)

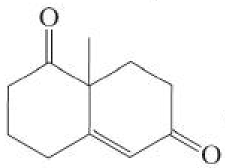
a. 

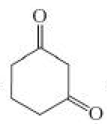
b. 

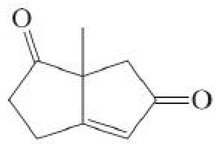
39. Based on the mechanism presented for the Pd-catalyzed reaction of a halobenzene with hydroxide ion (Section 22-4), write out a reasonable mechanism for the Pd-catalyzed reaction of Problem 38, part (a).

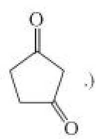
40. Using the methods described in this chapter, design a multistep synthesis of each of the following molecules, making use of the indicated building blocks as the sources of all the carbon atoms in your final product.

a.  , from $CH_3CO_2CH_2CH_3$ and $CH_3\overset{O}{\overset{\|}{C}}CH{=}CH_2$

b.  , from CH_3I, $CH_2(CO_2CH_2CH_3)_2$ and

$CH_3\overset{O}{\overset{\|}{C}}CH{=}CH_2$ (**Hint:** First make  .)

c.  , from CH_3I, $CH_2(CO_2CH_2CH_3)_2$ and

$BrCH_2\overset{O}{\overset{\|}{C}}CH_3$ (**Hint:** First make  .)

41. Give the products of reaction of the following aldehydes with catalytic *N*-dodecylthiazolium bromide. **(a)** (CH3)2CHCHO;$(CH_3)_2CHCHO$; **(b)** C6H5CHO;C_6H_5CHO; **(c)** cyclohexanecarbaldehyde; **(d)** C6H5CH2CHO.$C_6H_5CH_2CHO$.

42. Give the products of the following reactions.

a. C6H5CHO+HS(CH2)3SH→BF3$C_6H_5CHO + HS(CH_2)_3SH \xrightarrow{BF_3}$

b. Product of (a)+CH3CH2CH2CH2Li→THF

$\text{Product of (a)} + CH_3CH_2CH_2CH_2Li \xrightarrow{THF}$

What are the results of reaction of the substance formed in (b) with each aldehyde in Problem 41, followed by hydrolysis in the presence of HgCl2?$HgCl_2$?

43. **(a)** On the basis of the following data, identify unknowns A, found in fresh cream prior to churning, and B, possessor of the characteristic yellow color and buttery odor of butter.

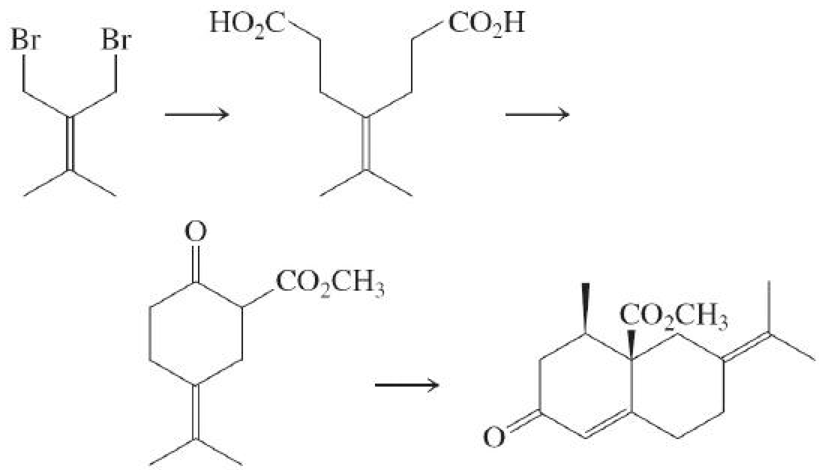
A: MS: m/z (relative abundance)=88(M+, weak), $\mathrm{MS}: m/z$ (relative abundance) $= 88(\mathrm{M}^{+}$, weak), 45(100), $45(100)$, 43(80). $43(80)$.
1H NMR: δ=1.36 (d, J=7 Hz, 3 H), ${}^{1}\mathrm{H}$ NMR : $\delta = 1.36$ (d, $J = 7$ Hz, 3 H),
3.73 (broad s, 1 H), 3.73 (broad s, 1 H), 4.22 (q, J=7 Hz, 1 H) ppm. 4.22 (q, $J = 7$ Hz, 1 H) ppm.
13C NMR: δ=19.5, ${}^{13}\mathrm{C}$ NMR : $\delta = 19.5$, 24.9, 24.9, 73.2, 73.2, 211.1 ppm. 211.1 ppm.
IR: v˜=1718 and 3430 cm-1. IR : $\tilde{v} = 1718$ and 3430 cm^{-1}.

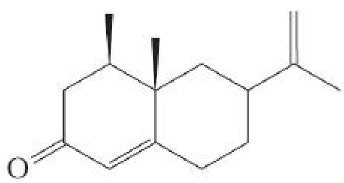
B: MS: m/z (relative abundance)=86(17), $\mathrm{MS}: m/z$ (relative abundance) $= 86(17)$, 43(100). 43(100).
1H NMR: δ=2.29 (s) ppm. ${}^{1}\mathrm{H}$ NMR : $\delta = 2.29$ (s) ppm.
13C NMR: δ=23.3, 197.1 ppm ${}^{13}\mathrm{C}$ NMR : $\delta = 23.3$, 197.1 ppm
IR: v˜=1708 cm-1. IR : $\tilde{v} = 1708$ cm^{-1}.

(b) What kind of reaction is the conversion of compound A into compound B? Does it make sense that this should take place in the churning of cream to make butter? Explain. **(c)** Outline laboratory syntheses of A and B, starting with molecules containing only two carbons. **(d)** The UV spectrum of A has a λmax=271 nm, $\lambda_{\max} = 271$ nm, whereas that of B has a λmax=290 nm. $\lambda_{\max} = 290$ nm. [Tailing of the latter absorption into the visible region of the spectrum (Section 14-11) is responsible for the yellow color of B.] Explain the difference in λmax. $\lambda_{\max}$.

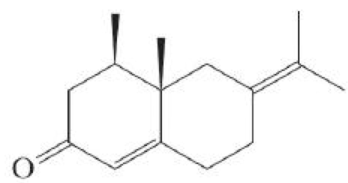
44. Write chemical equations to illustrate all primary reaction steps that can occur between a base, such as ethoxide ion, and a carbonyl compound, such as acetaldehyde. Explain why the carbonyl carbon is not deprotonated to any appreciable extent in this system.

45. The nootkatones are bicyclic ketones that contribute to the flavor and aroma of grapefruits [in addition to 2-(4-methyl-3-cyclohexenyl)-2-propanethiol, Section 9-11]. Nootkatone is also an environmentally friendly insect repellent against ticks, mosquitos, and termites. Fill in the missing reagents in the partial synthetic sequence for the preparation of isonootkatone shown below. More than one synthetic step may be needed for each transformation.



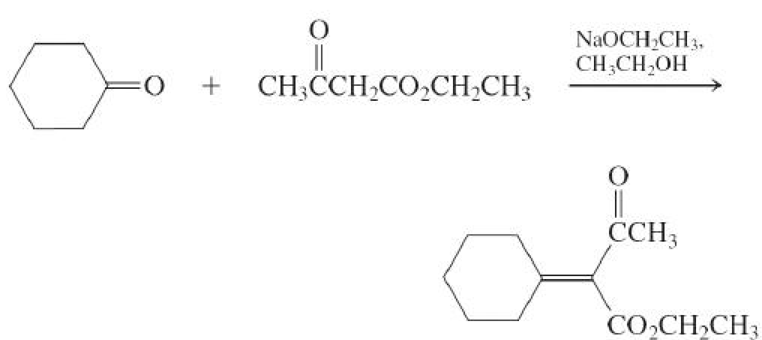


Nootkatone

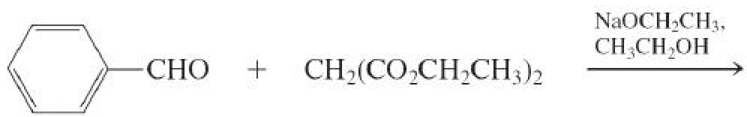


Isonootkatone

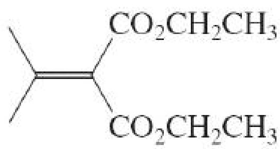
46. **β**β-Dicarbonyl compounds condense with aldehydes and ketones that do not undergo self-aldol reaction. The products are αα,ββ-unsaturated dicarbonyl derivatives, and the process goes by the colorful name of Knoevenagel condensation. **(a)** An example of a Knoevenagel condensation is given below. Propose a mechanism.



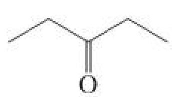
(b) Give the product of the Knoevanagel condensation shown below.

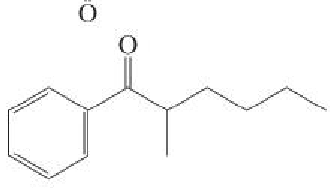


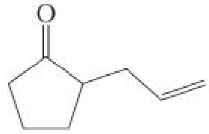
(c) The diester shown below is the starting material for the dibromide used in the synthesis of isonootkatone (Problem 45). Suggest a preparation of this diester using the Knoevenagel condensation. Propose a sequence to convert it into the dibromide of Problem 45.



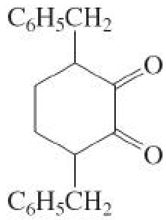
47. The following ketones cannot be synthesized by the acetoacetic ester method (why?), but they can be prepared by a modified version of it. The modification involves the preparation (by Claisen condensation) and use of an appropriate 3-ketoester, $R\overset{O}{\overset{\|}{C}}CH_2\overset{O}{\overset{\|}{C}}OCH_2CH_3$, containing an R group that appears in the final product. Synthesize each of the following ketones. For each, show the structure and synthesis of the necessary 3-ketoester as well.

a. 

b. 

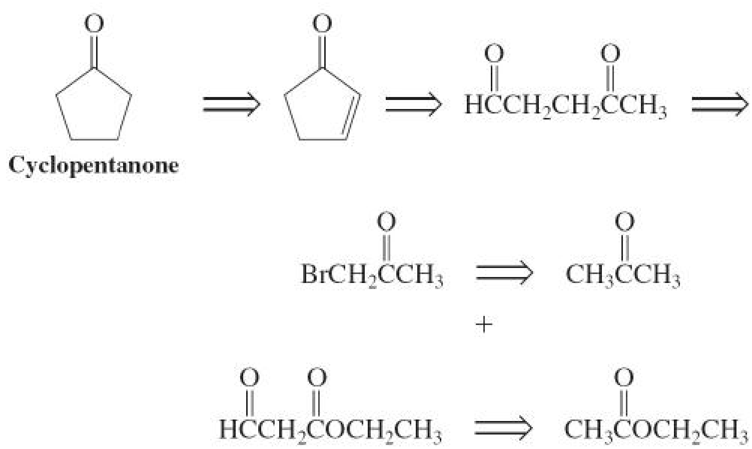
c. 

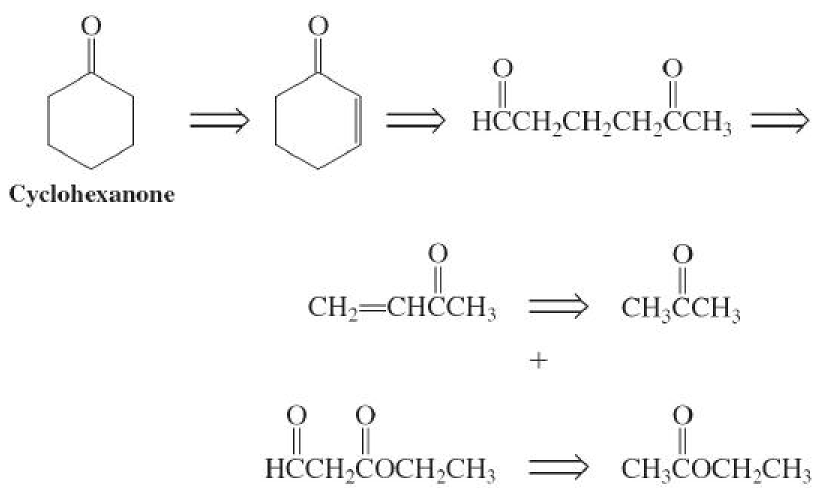
(**Hint:** Use a Dieckmann condensation.)

d. 

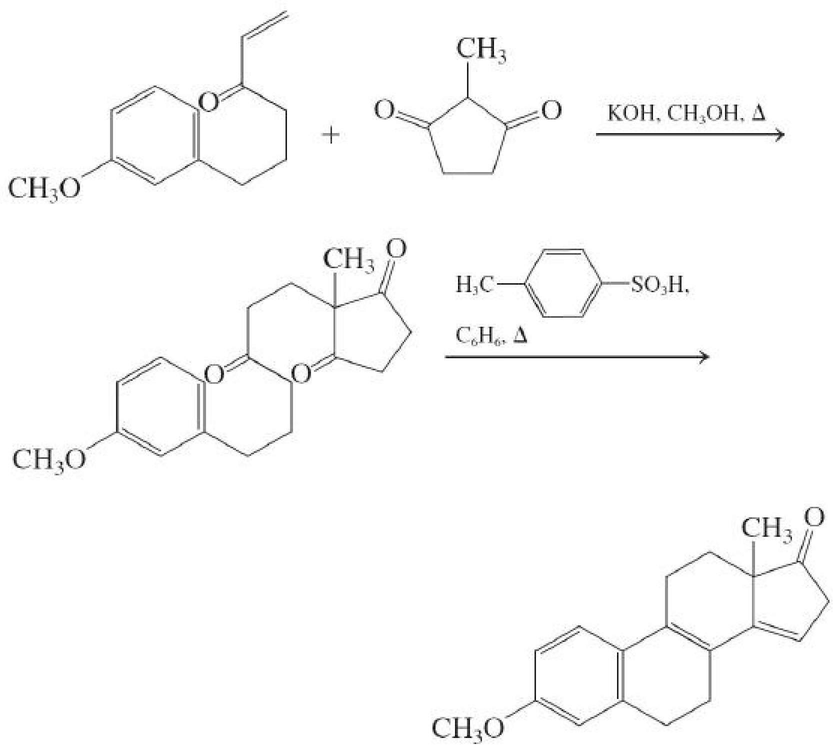
(**Hint:** Use a double Claisen condensation.)

48. Some of the most important building blocks for synthesis are very simple molecules. Although cyclopentanone and cyclohexanone are readily available commercially, an understanding of how they can be made from simpler molecules is instructive. The following are possible retrosynthetic analyses (Section 8-8) for both of these ketones. Using them as a guide, write out a synthesis of each ketone from the indicated starting materials.

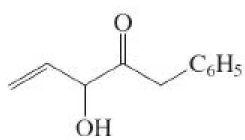


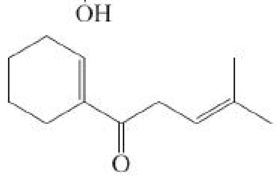


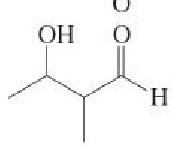
49. A short construction of the steroid skeleton (part of a total synthesis of the hormone estrone) is shown here. Formulate mechanisms for each of the steps. (**Hint:** A process similar to that taking place in the second step is presented in Problem 46 of Chapter 18.)



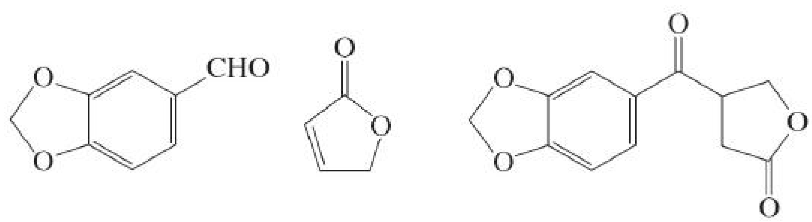
50. **CHALLENGE** Using methods described in Section 23-4 (i.e., reverse polarization), propose a simple synthesis of each of the following molecules.

a. 

b. 

c. 

51. **CHALLENGE** Propose a synthesis of ketone C, which was central in attempts to synthesize several antitumor agents. Start with aldehyde A, lactone B, and anything else you need.

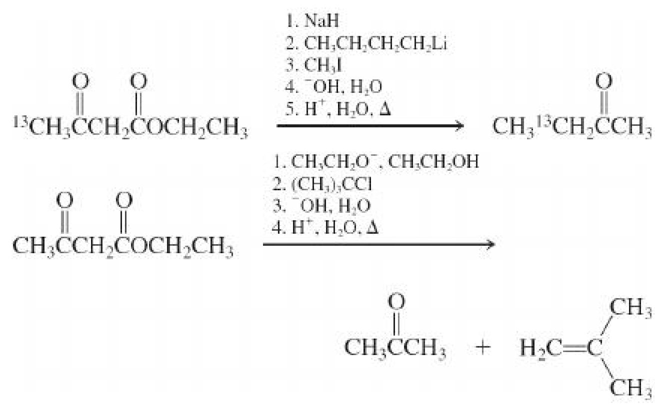


A B C

Team Problem

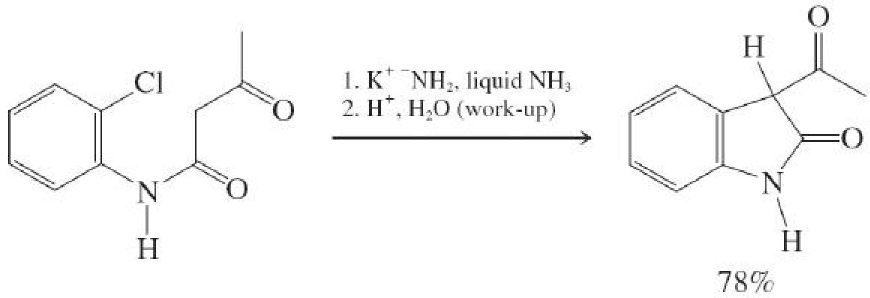
52. Split your team in two, each group to analyze one of the following reaction sequences by a mechanism (13C=carbon-13 isotope). (^{13}C = carbon-13 isotope).

a. b.



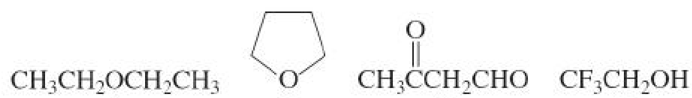
Reconvene and discuss your results. Specifically, address the position of the 13C^{13}C label in the product of (a) and the failure to obtain alkylation in (b).

As a complete team, also discuss the mechanism of the following transformation. (**Hint:** For the first step, a minimum of three equivalents of KNH2KNH_2 is required.)



Preprofessional Problems

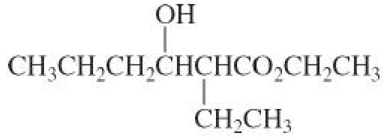
53. Two of the following four substances are more acidic than CH3OH $CH_3\ OH$ (i.e., two of these have KaK_a greater than methanol). Which ones?

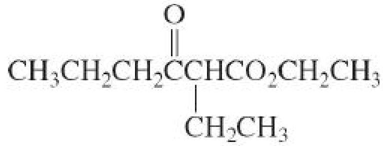


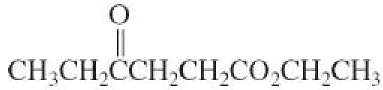
A B C D

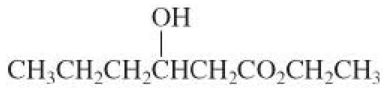
- **a.** A and B;
- **b.** B and C;
- **c.** C and D;
- **d.** D and A;
- **e.** D and B.

54. The reaction of ethyl butanoate with sodium ethoxide in CH3CH2OH CH_3CH_2OH gives

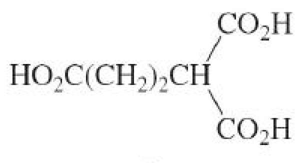
a. 

b. 

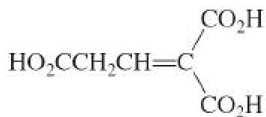
c. 

d. 

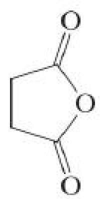
55. When acid A (below) is heated to $230^\circ C$, CO_2 and H_2O are evolved and a new compound is formed. Which one?



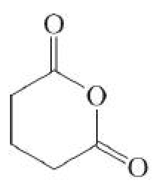
A

a. 

b. $HO_2CCH_2CH_2CH_2CH_3$

c. 

d. $CH_3CH_2CH(CO_2H)_2$

e. 

56. A compound with m.p. $-22^\circ C$ has a parent peak in its mass spectrum at $m/z = 113$. The ^1H NMR spectrum shows absorptions at $\delta = 1.2$ (t, 3 H), 3.5 (s, 2 H), and 4.2 (q, 2 H) ppm. The IR spectrum exhibits significant bands at $\tilde{\nu} = 1750$, 2250, and $3000\ cm^{-1}$. What is its structure?